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Final Proposal
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Transit-Oriented Development Mixed-Use Zones (TODX)

Densifying MTA subway station areas with incentives to achieve neighborhood-wide target densities, and the inclusion of flexible mixes of use.

Abstract:

This proposal will address the need for increased density near New York City transit stations that are not well used. Although the MTA is currently well-funded, in the past, many stations were at risk to close due to extensive capital backlogs and low ridership. Increasing density near stations, and ridership at stations, will reinforce the subway as the backbone of the community, and will make it immune to any future political meddling or fiscal shortfalls. Additionally, many of these stations have non-efficient land uses in their proximity. Amidst our current affordability crisis, this untapped and underutilized land allows us to achieve growth in a more efficient and sustainable way.

To develop a solution to this problem, I have taken inspiration from London, England, and Vancouver, Canada. New York's subway is very vast, with 472 stations. Without a method of evaluation and prioritization, it would be difficult to begin to make progress. London has frameworks and plans where they evaluate transit-access, opportunity zones, and set targets for densification and development that I will draw inspiration from. Additionally, much of the land near underused stations in New York has current commercial and industrial uses, and Vancouver provides an example for densifying industrial areas near transit with mixed-uses, while still maintaining the industrial uses that pre-exist the change.

To achieve these goals, I propose a new Transit Oriented Development Mixed-Use Zone (TODX). TODX zones will set targets for density in radial zones around the transit station, calculated by dividing the total built area of the zone by the total lot area of the zone, and adjusting based on station type (local, express, transfer), and by lot size. The main mechanism for growth in this zone will be FAR bonuses with intensities relative to the distance from the

station, that will exist until the desired Station-Area Density is reached. There will be additional requirements around form, bulk, affordability, and use to participate in the program.

Case Study #1: London

In London, Transport for London (TfL) is the planner and operator of public transit. They operate the underground (metro), buses, light-rail, overground (regional rail/metro hybrid) and more. In their planning process, they assess the connectivity of the city to determine ideal locations for transportation improvements, community resources, suitable locations for housing and commercial development, and to determine if areas need changes to parking supply.

They use three methods to assess the connectivity of the city: Public Transport Access Level (PTAL), which assigns a rating to areas based on transit proximity and the level of service provided; travel time mapping, to graph how long it takes to travel across the city; and catchment analysis, to understand what services and jobs exist within a certain distance to specific areas. Travel time mapping is intriguing, but it is most effective at conveying connectivity relative to specific areas, rather than the whole system. Catchment analysis depends on the existing built form and land use of surrounding areas and therefore isn't applicable[1].

Therefore, will look specifically at PTAL, as I am interested at understanding how well connected a specific part of the mass transit system is to the whole. With this, I will be able to determine how much density and development would be rational given a specific station-area. Below is a bulleted list of key factors that influence how an area is scored in London. Important to note is that PTAL doesn't reflect the level of crowding at the stations or on board the trains.

1. Transport for London. *Accessing Transport Connectivity in London, 4*

Public Transport Access Level Key Factors

- Walking distance to nearest station
- Wait time at said station
- Number of services at nearest station
- Proximity to Major station

PTAL is used in numerous ways within urban planning in London. It is a major factor in determining the ideal density of housing in parts of the city, and this is reflected in the city's comprehensive plan, the London Plan. PTAL is also used to recommend parking quantities, with good access areas having less parking, while still acknowledging parts of the city without good connectivity to public transit will still need parking. Additionally, PTAL is tracked within commercial areas, with goals set to have most of the commercial space in high PTAL areas[2].

Setting	Public Transport Accessibility Level (PTAL)		
	0 to 1	2 to 3	4 to 6
Suburban	**150-200 hr/ha**	**150-250 hr/ha**	**200-350 hr/ha**
3.8-4.6 hr/unit	35-55 u/ha	35-65 u/ha	45-90 u/ha
3.1-3.7 hr/unit	40-65 u/ha	40-80 u/ha	55-115 u/ha
2.7-3.0 hr/unit	50-75 u/ha	50-95 u/ha	70-130 u/ha
Urban	**150-250 hr/ha**	**200-450 hr/ha**	**200-700 hr/ha**
3.8-4.6 hr/unit	35-65 u/ha	45-120 u/ha	45-185 u/ha
3.1-3.7 hr/unit	40-80 u/ha	55-145 u/ha	55-225 u/ha
2.7-3.0 hr/unit	50-95 u/ha	70-170 u/ha	70-260 u/ha
Central	**150-300 hr/ha**	**300-650 hr/ha**	**650-1100 hr/ha**
3.8-4.6 hr/unit	35-80 u/ha	65-170 u/ha	140-290 u/ha
3.1-3.7 hr/unit	40-100 u/ha	80-210 u/ha	175-355 u/ha
2.7-3.0 hr/unit	50-110 u/ha	100-240 u/ha	215-405 u/ha

Figure 1. Housing density recommendations from the London Plan. Courtesy of Transport for London

2. Transport for London. *Accessing Transport Connectivity in London, 6-7*

Above is a figure that dictates the housing density goals based on PTAL. In the table, hr stands for habitable rooms, u is unit, and ha is hectare. It is very clear how different access levels enable for more dense housing. Additionally, it is interesting how the plan lays out different prescriptions of density to account for parts of the city where multi-bedroom homes may be more or less prevalent[3].

After identifying the PTAL, I searched to see if New York could meet similar criteria that would make such a metric useful. According to C40, an organization representing global large cities (mainly focused on climate), PTAL would be useful for a city trying to reduce parking provisions and increase developable space. In the US, only a few cities other than New York have such the dilemma of vast public transit systems with undeveloped adjacent land. In discussing success in London, C40 attributes it to its citywide nature, its transparent methodology, and the accessibility of information to planners[4].

In addition to a well-defined metric, London also designates Opportunity Areas. Two factors that led to Opportunity Areas are large projections of future population growth, as well as central government threats to withdraw grants for TfL. This situation is similar to the MTA's, where despite a good fiscal situation now, there is long-term unpredictability. In London, this has led to the prioritization of high-density development to subsidize public transit, as well as provide necessary ridership. This was seen in the first London Plan (2004) where Opportunity Areas were first defined (approx. 50 today). Opportunity areas are spread across different parts of the city, and are frequently located near transit that contains brownfield, suburban and industrial uses[5].

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3. Transport for London. *Accessing Transport Connectivity in London, 6-7*
4. C40 Cities, *C40 Good Practice Guides.*
5. Ferm et al., *Documenting the 'Soft Spaces' of London Planning., 396*

In looking at solutions for New York, both PTAL and Opportunity Areas can be used to inform relative levels of zoned density near MTA stations. Doing so will allow for MTA stations to be used more efficiently, reducing their fiscal risk, and will also allow for significant amounts of development in areas that have good access to the rest of the city via the subway.

Case Study #2: Vancouver

Similarly to New York, Vancouver has rapid population growth, is one of its country's largest ports, and has limited land. The need for housing and the retention of industrial land is in great tension in both cities. We will look at the conditions in Vancouver, as well as some solutions they have prioritized to determine how to best consider the light industrial land that is in proximity to MTA subway stations.

Firstly, it is essential to differentiate between the differing types of industrial land uses, as some are much more intense and disruptive than others. Some types of industrial uses are suitable for multi-story industrial use, while others may not be. It is essential to provide a variety of typologies of industrial land throughout the city and not pigeonhole all uses into one typology. In Vancouver, this is covered by their four-step plan for the future of industrial areas: 1) Protect Industrial Land to avoid net loss of space; 2) Modernize zoning to allow and encourage multi-story zoning when possible; 3) Match specific industrial uses to the right spaces; 4) Monitor and report industrial change. This plan gives industrial uses a safety net, while providing opportunities for growth and expansion[6].

Not all multi story industrial should be exclusively industrial. However, it is important to prioritize industrial uses in these cases, as it is easy for commercial and residential space to take over given their ability to pay higher rents. One example of a zone is I-1, which allows 3 FAR (1

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6. Aderneck, Eric. *Industrial Lands Densification and Intensification, 75*

for industrial), with the industrial use required on the ground floor. This zone is meant to be located near transit and should be adjacent to residential and commercial zones. Within the zone, uses compatible or complementary to light industry are allowed. In some cases, the FAR was increased, but industrial space is still 33% of the total floor area[7].

There is also an I-2 zone, which is meant for industrial uses that don't mix well with residential areas. In these spaces, it could work to have commercial, production, labs and office spaces in addition to the industrial uses. Height limits in these areas must account for the high floor heights that are required in industrial buildings. Additionally, consideration should be given to e-commerce warehouses and last mile delivery, to ensure they are permitted under the new zoning[8].

Beyond just defining zones, Vancouver has done much research to understand the relationships between different types of industrial uses and non-industrial uses. In figure 2 below, we can understand the types of industrial uses that would work in Vancouver's I-1 residential adjacent zones. These types will be compatible within the New York proposal. Due to the transit proximity, density and small radius of zone, all industrial uses in these zones must be compatible with residential.

Some uses that are compatible with residential in Vancouver include, food and beverage manufacturing, textiles and clothing manufacturing, storage and distribution centers, and even more medium intensity uses such as IT and communications[9]. Beyond just uses, Vancouver also explores the relationship between industrial and residential uses, and noise. Considerations are made both for locational relationships to noise, as well as noise mitigation requirements[10].

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7. Aderneck, Eric. *Industrial Lands Densification and Intensification, 78-79*
8. Aderneck, Eric. *Industrial Lands Densification and Intensification, 79-80*
9. Ng, Victoria. *Mitigating Industrial Land Use Impact, 60*
10. Ng, Victoria. *Mitigating Industrial Land Use Impact, 25*

			Light				Medium				Heavy		
			Light Manufacturing		Warehouse	Other	Manufacturing (Material and Products)				Miscellaneous		
Category	**Description**	**Land Use**	Food/Beverage Products Manufacturing, Coffee Roasters	Clothing, Creative Products, Jewellery, Textile or knit goods	Storage, Wholesale, Distribution Centre	Brewing/ Distilling, Bakery products	Chemical Products, Biotech	Furniture/Fixtures, Wood products, Pulp, rubber, Shoes/Boots, Leather, Paper, Plastics, Wood, Miscellaneous	Information Communication Technology, 3D Printing, VR, e-commerce	Linoleum, non-metallic mineral, machinery/ equipment, petroleum/coal, transportation equipment, metals, motor vehicle parts	Casino	Data Centre *	Aircraft landing place, Junk Yard/Shop, Cardlock fuel station, Booming Ground
Residential	Residential/ Dwelling Unit	D	Compatible*	Compatible*	Compatible*	Incompatible	Noxious	Noxious	Compatible*	Noxious	Noxious	Noxious	Noxious
Commercial	Artist Studio	C	Incompatible	Compatible	Compatible	Compatible*	Noxious	Compatible*	Compatible*	Incompatible	Noxious	Noxious	Noxious
	Bingo Hall	C	Incompatible	Compatible*	Compatible*	Compatible*	Incompatible	Noxious	Compatible	Noxious	Compatible	Noxious	Noxious
	Fitness Centre	C	Compatible*	Compatible	Compatible	Compatible*	Compatible*	Incompatible	Compatible	Noxious	Compatible	Noxious	Noxious
	Theatre	C	Noxious	Compatible*	Incompatible	Noxious	Noxious	Noxious	Incompatible	Noxious	Compatible*	Noxious	Noxious
	Sign Painting Shop	C	Incompatible	Compatible	Compatible	Incompatible	Compatible	Compatible	Compatible	Noxious	Incompatible	Noxious	Noxious
Institutional	Church	I	Incompatible	Compatible	Compatible	Incompatible	Compatible*	Noxious	Compatible*	Noxious	Noxious	Noxious	Noxious
Office	Design Offices	n/a	Compatible	Compatible	Compatible	Compatible	Compatible*	Compatible*	Compatible	Noxious	Noxious	Noxious	Noxious
	Info., Comm., Tech., Manufact. Offices	n/a	Compatible*	Compatible*	Compatible	Compatible	Compatible*	Compatible*	Compatible	Noxious	Noxious	Compatible*	Noxious
	General Office	O	Compatible	Compatible*	Compatible	Compatible	Compatible	Incompatible	Compatible*	Noxious	Compatible*	Noxious	Noxious
	Financial Institution	O	Compatible	Compatible*	Compatible	Noxious	Incompatible	Incompatible	Compatible*	Noxious	Incompatible	Noxious	Noxious
	Health Care Office	O	Compatible	Compatible*	Compatible	Incompatible	Compatible*	Incompatible	Compatible*	Noxious	Noxious	Noxious	Noxious
	Health Enhancement Centre	O	Compatible	Incompatible	Compatible	Incompatible	Compatible*	Incompatible	Incompatible	Noxious	Noxious	Noxious	Noxious
	Cannabis Store	R	Noxious	Compatible*	Compatible	Incompatible	Compatible*	Incompatible	Compatible	Noxious	Incompatible	Noxious	Noxious

Figure 2. Industrial Compatibility Matrix from the City of Vancouver[11]

In locations where both industrial uses and residential uses are nearby, there are requirements for the volume of machinery and or activity. They also recommend co-locating uses whose time of use do not overlap, for example a manufacturing facility that is busy during the day, and a concert hall which is busy during the evening. Furthermore, in instances where development occurs next to preexisting industrial (or highway, rail transit) areas, they have three different guidance: 1) minimize doors and windows along the noise-prone areas; 2) use sound absorbing materials, such as carpets, ceiling tiles, special insulation; 3) consider that noise travels through the building structure, build accordingly, and isolate floor noises. Lastly, they recommend that noise-proof areas are built to the full height of the noise source[12]. In the case of transit parallel, the noise proof base would need to match the height to be effective.

Turning back to New York, there is a lot we can adapt from Vancouver in order to

11. Ng, Victoria. *Mitigating Industrial Land Use Impact, 60*
12. Ng, Victoria. *Mitigating Industrial Land Use Impact, 29*

successfully densify our transit adjacent industrial areas. We must be sure to retain our existing industrial lands, as they are becoming rarer, yet are essential to a city's productivity. We also must consider the type of industrial use and its compatibility to a high-density residential environment. Lastly, we will consider the spatial relationship between the uses and noise-emitting activities to inform bulk and form requirements.

Proposal: Transit-oriented density envelopes

To achieve goals of densification near underused MTA stations, I propose a new zone, the Transit Oriented Development Mixed-Use Zone, otherwise known as TODX. There are two main interventions that this zone will provide, which are: 1) Incentivizing densification up to specific density targets near MTA stations; and 2) Allowing stations to become mixed-use centers without losing industrial land. With these interventions stated, I will first discuss the criteria for site selection.

Sites of Intervention

Taking inspiration from London's PTAL, I will introduce transit-oriented zones around stations based on the station's connectivity relative to the rest of the subway system. The first way to classify station areas is by the connectivity of the station itself. Stations will be grouped into three groups: **Local** stations, **Express** stations, and **Transfer** stations. Although this is simpler than London's evaluation of number of services, service frequency and if it is a "Major" rail stop, this categorization will serve well for New York.

Local stations typically have less services, and due to this, have longer headways when compared to express lines. Currently, these station areas sometimes have density, but we must be sensitive not to over-densify them since commutes often require transferring or long rides.

Express stations have more than one service and allow passengers greater connectivity by skipping local stops. Currently, these stations are often more commercial than their local counterparts, but there is still additional capacity for density beyond the main commercial corridor. These stations will have headways that are typically 3 minutes or less during rush hour and will be able to handle significant growth if they are currently underused.

Transfer stations are express stations (except for a few edge cases) that connect multiple lines. Not only are passengers well connected along one line but also are well connected in the orthogonal direction. These stations need to be incredibly dense, not only to allow its residents and office-goers to access wide amounts of the city easily, but also to allow people from across the city to be able to reach services in an extremely well-connected area.

With this hierarchy stated, we will] determine a way to measure station area density. An effective way to do this is by summing the total lot area within a quarter mile (about a 10-minute walk) of the station, the total built floor area within a quarter mile of the station, and then dividing the total built floor area by the total lot area to calculate the Floor Area Ratio (FAR) of the entire neighborhood. With this calculation, we can calculate what existing station areas station-area FAR's are to get a picture of the meaning of the value.

Station Name and Services	**Station Classification**	**¼ mile Station-Area FAR**
79 St, (D)	Local Station	1.15
Carroll St (F)	Local Station	1.69
59 St (N, R)	Express Station	1.75
Metropolitan St – Lorimer St (G, L)	Transfer Station	2.23
Delancey St – Essex St (F, J, M, Z)	Transfer Station	4

Table 1. Example stations and their Station-Area FARs

Station Density Targets

From these values, we can understand a few things. Firstly, the hierarchy of local, express and transfer stations is true in the sample of stations. Secondly, we can use these values to guide targets for each category. In guiding targets, I will also consider walking proximity to the station, as done in London. This will result in an outer quarter mile radius zone around the station, and an inner eighth mile radius zone. All inner zones will have a target neighborhood FAR that is 0.5 higher than the quarter mile zone.

In the case of Local stations, 79 St is mostly single family or small multi-family buildings. A rider at this station can reach Downtown Brooklyn in 20 minutes, or Midtown Manhattan in about 45 minutes. Although it is far into Brooklyn, there is no reason that houses must be detached, or that moderately sized apartment buildings shouldn't be present. When compared with Carroll St the difference can be seen. This area is densely built with row homes (single and multi-family), with the occasional apartment building. For a local station, this typology makes sense. There is no need to build super tall buildings to achieve a dense urban fabric. However, if for whatever reason a building had to be torn down, it would be acceptable if a slightly larger one would replace it. For this reason, the quarter mile density target for local stations is 1.75.

Presently, Express stations are often built in a way I would consider desirable for Local stations. This means that they have dense use and are well suited for decent transit ridership, but their built environments do not reflect that they are extremely well-connected areas. Looking at 59 St in Brooklyn, riders can access Midtown Manhattan in about 30 minutes. Additionally, they can take the local train that enables them to also reach Lower Manhattan. With this amount of

connectivity, the station area should contain more than row homes and a few apartment buildings with similar heights. Therefore, for express stations the quarter mile density target should be 2.5.

Lastly, Transfer stations have immense opportunity for density. Looking at Delancey St – Essex St, we can understand what a station-area FAR of 4 would be. This area has many new tall mixed-use buildings but also has plenty of old-style row tenements. Additionally, the abandoned Essex Market takes up nearly a square block and is only one or two floors. In recommending a density for transfer stations, I will set a less intense target than this station. Firstly, it is in Manhattan, and most of the stations I will address are in the outer boroughs. Secondly, I just don't think this much density is necessary. If every transfer station grows to this, that would be wonderful, but setting this as a goal for all stations may be too much. That being said, we will set the quarter mile density at 3, specifically so the eighth of a mile density is 3.5, which is still less dense than Delancey St.

Table 2: Target Densities of Station Areas

Station Type	**Quarter-mile radius target**	**Eighth-mile radius target**
Local	1.75	2.25
Express	2.5	3
Transfer	3	3.5

Incentives

Having defined the target densities of different areas, I will now explain the incentive structure. To achieve target densities, every lot in the quarter mile radius will be eligible for a **25% FAR bonus**, and lots within an eighth of a mile will be eligible for a **50% FAR bonus.** These bonuses will expire when the total neighborhood density hits the targets as calculated

above. Additionally, if the eighth mile target is hit before the quarter mile, the eighth mile incentive will expire, but the quarter mile will remain for the whole area. Likewise, if the quarter mile density is achieved first, its incentive will expire but the eighth mile will remain.

In addition to the quarter mile and eighth mile zones, we will draw a half mile zone around all stations that will get very modest incentives. By drawing half mile areas, we can connect all the station nodes along the line, allowing transit-oriented density to continue along the corridor of the entire route. The incentive in these areas will be a **10% FAR bonus**, and it will expire when the quarter mile density is achieved. Unlike the other zones which have their zones essentially rewritten, this bonus will be an overlay on the existing zoning of these areas.

Requirements to obtain incentive

For an individual lot to take advantage of these FAR bonuses, they must follow criteria that are defined both by the zone they are in and their individual lot sizes. Lots that are larger than 10,000 sq. ft. (large lots) that are in the quarter-mile zone must consist of residential uses on all but their two bottom-most floors, where they are then required to have a non-residential use. For this space, one of the following public interest uses must be provided: grocery store, childcare, pharmacy/healthcare, school, community center, recreation center, library or post office. These spaces may also be used to create publicly accessible parks, so long as the open space is no more than one quarter of the lot. If this option is chosen, buildings will not be given additional height to work with. It is permissible to sub-divide the space and allow other uses, so long as 50% of the commercial space is an above use. Inside the eighth mile zone, the requirement remains as the bottom two floors, but allows up to 50% commercial use. Additionally, if the lot faces an elevated transit line, the building can exclude residential uses, and is required to have a street wall that is fifteen feet above the track height for sound insulation.

Lots that are not “large” have the option to have commercial uses on their ground floor only and can provide uses such as the public-interest ones, or coffee shops, restaurants, bars, boutiques and similar uses. There is no requirement here, but such lots will have the opportunity to deduct half of the commercial ground floor FAR from the building’s total calculation in the quarter mile area, and the total commercial FAR in the eighth mile area. However, these lots will be subject to “sliver building” requirements, ensuring that they are no taller than the width of the street or 100 feet if they are less than 45 feet wide.

For commercial and industrial lots that may have significantly lower original FAR’s than their neighboring residential uses, they may calculate the average FAR of all adjacent lots to set a new baseline FAR for the lot. The same requirements will stand for distribution of uses, except for the case of lots that were formerly manufacturing. For these lots, there is a requirement to provide the same industrial floor area that existed prior to the development. In the case that this occupies more than two floors in the quarter-mile radius, this is permitted. The permitted uses for these areas will be like Vancouver’s, including food and beverage manufacturing, clothing and textiles, small warehouses, micro-hubs and distribution centers and non-nuisance IT uses.

The methodology behind these requirements is that large lots will provide spaces for necessary community amenities and assets, while smaller lots will be able to provide infill commercial space at varying price ranges that will make the neighborhood diverse and vibrant. Specifically for smaller lots, they would be able to convert ground floors to these uses without having to rebuild the entire building! Given the possibility of light industrial uses near residential uses, we will ensure there is no net loss in industrial floor area.

Beyond these requirements, there will also be an affordability requirement. All demolished affordable units must be replaced 1:1, and additionally, for all incentive FAR, 50%

to 60% of the units must be affordable. Requirements will be like MIH, where 50% of floor area would map to option 3 of MIH, 55% would map to option 1, and 60% would map to options 2 and 4. They must be distributed throughout the building, on all floors, with access to the same amenities as all other residents. They also must meet bedroom mix requirements. This will ensure that as our subway station areas densify, the resulting dwellings are accessible to people of all incomes.

Table 3: Requirements for different lot sizes and distances from station

Building Type	**Incentive (FAR Bonus)**	**Residential Usage**	**Commercial/Industrial Usage**	**Permitted Usages**
Large Lot in Quarter Mile	25%	All floors but two bottom-most, incentive FAR must be 50-60% affordable	Two bottom-most floors required	**Required** on 50% commercial: grocery store, childcare, pharmacy/healthcare, school, community center, recreation center, library or post office. **Industrial if prev. use.** **Remainder**: store, chain store, restaurant, bar, boutique, barber shop, laundry, etc.
Large Lot in Eighth Mile	50%	50% to all but 2 floors. Same afford. as above	Two bottom-most floors required, 50% max.	Same as above ^
Small Lot in Quarter Mile	25%	All but lowest floor. Same afford. as above	Optional ground floor, half of FAR may be deducted	Boutique, restaurant, bar, barber shop, laundry, etc.
Small Lot in Eighth Mile	50%	All but lowest floor. Same afford. as above	Optional ground floor, total FAR may be deducted	Same as above ^

Additional TODX attributes

Because we are densifying with the goal of transit ridership, we will greatly change parking requirements to shift modes and improve the pedestrian experience. Inside of the half mile zone, we will eliminate parking minimums. Within the quarter mile zone, we will ban surface parking and introduce parking maximums of 1 per 4 dwelling unit residential, and 1 per 1000 sq. ft. commercial use. Within the eighth mile zone, we will ban parking entirely.

Next, we will consider dwelling density. We will remove the dwelling unit factor in both the quarter mile and eighth mile zones. This will allow developers to adequately meet the demand for units that is present from prospective tenants. Additionally, this will allow preexisting buildings to sub-divide into smaller units so long as they meet the requirements of the building code. This will help to enable smaller, more affordable units, allow owners to create a second unit to help them continue to afford ownership, and will allow more households to live within greater proximity to transit.

There will also be a minimum unit count, which will ensure that development is occurring for buildings of all types, not just luxury buildings. The calculation for this will be the total residential floor area divided by 900, which is around the average size of a two-bedroom apartment. This will allow flexibility for family-sized units, while still providing studios and one-bedroom units as well.

Finally, I will address bulk regulations within the TODX zones. Side yard requirements will be removed, allowing for the building of row houses and the removal of driveways. Additionally, to build a continuous street wall, lots will be required to have their street-facing wall either within ten percent of the average of their adjacent buildings, or at a maximum of 10 feet from the lot line, whichever is less. This will require street walls in contexts that already

have them and will encourage the creation of street walls where they don't already exist. If a developer pursues a park as a use (as mentioned above) the street wall may not apply.

In terms of height, lots will multiply the existing height requirements by the FAR incentive (25 or 50 percent) to find their maximum heights. Base heights will remain the same as they already are. In cases where there is commercial and industrial land with drastically different requirements from its adjacent residential buildings, similarly to FAR they can calculate the average of the adjacent lots' base height and maximum height rules, and use those instead to create a consistent urban fabric. Lastly, in case these calculations do not provide this, there is a requirement to set back if the building becomes taller than the width of the street.

Station and lot eligibility

Now that we have defined the TODX zone, we must consider where it will be mapped. In London, they use PTAL to recommend development densities around all stations. However, they noted that this did not account for current levels of ridership and crowding. For that reason, I have decided that only stations in the lower half of subway station ridership will be included. This excludes most of Manhattan, which makes sense since the built environments of these areas mostly meet the recommendations already. However, there are certain stations with high ridership but low development density. These stations are often key transfer points for bus riders and commuter rail passengers. Due to the crowding at these stations already, we will avoid further densification. In *Figure 3*, the stations that will be included in the TODX mapping are those who are dark green and light green. Additionally, all the green dots below 59th St in Manhattan will be excluded, since these are often cases of poor station access, adjacent express stations, or poor stop spacing.

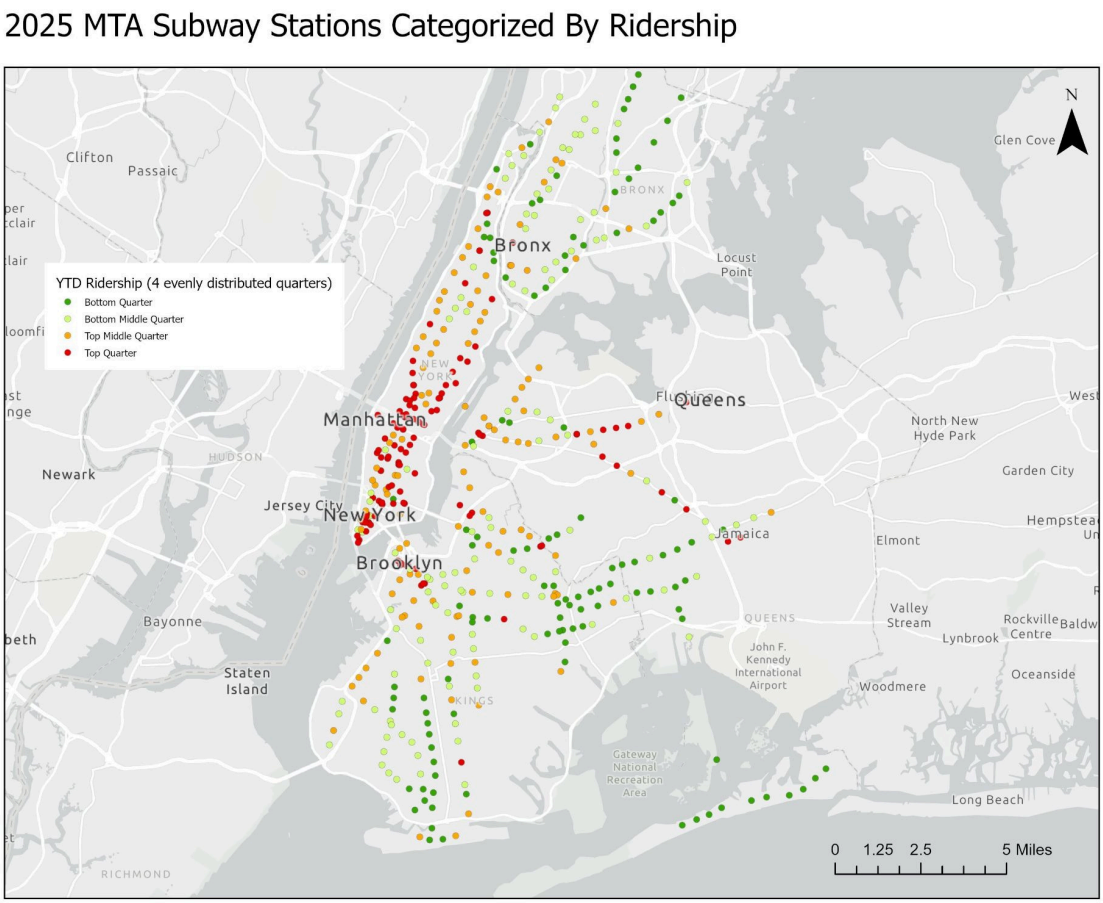


Figure 3. MTA Stations categorized by ridership.

Now that we have defined the per-station eligibility, we will define eligibility on a per lot level. For the quarter and eighth mile radius zones, if any part of the block is contained within the circle, the whole block will be included. By doing this, we will prevent fragmentation from occurring on a per-block level. In contrast, for the half mile zone, we will only include blocks that are entirely within the circle. A half mile walk to a transit station is still decently far, and if the entire block was considered, that could mean an additional 600-1000 foot walk beyond a half mile for those at the end of the block. It is simpler to restrict the half mile zone in this way. Below in *Figure 4* is a map of the included stations, with the quarter mile "outer" zone in yellow, and the eighth mile "inner" zone in orange.

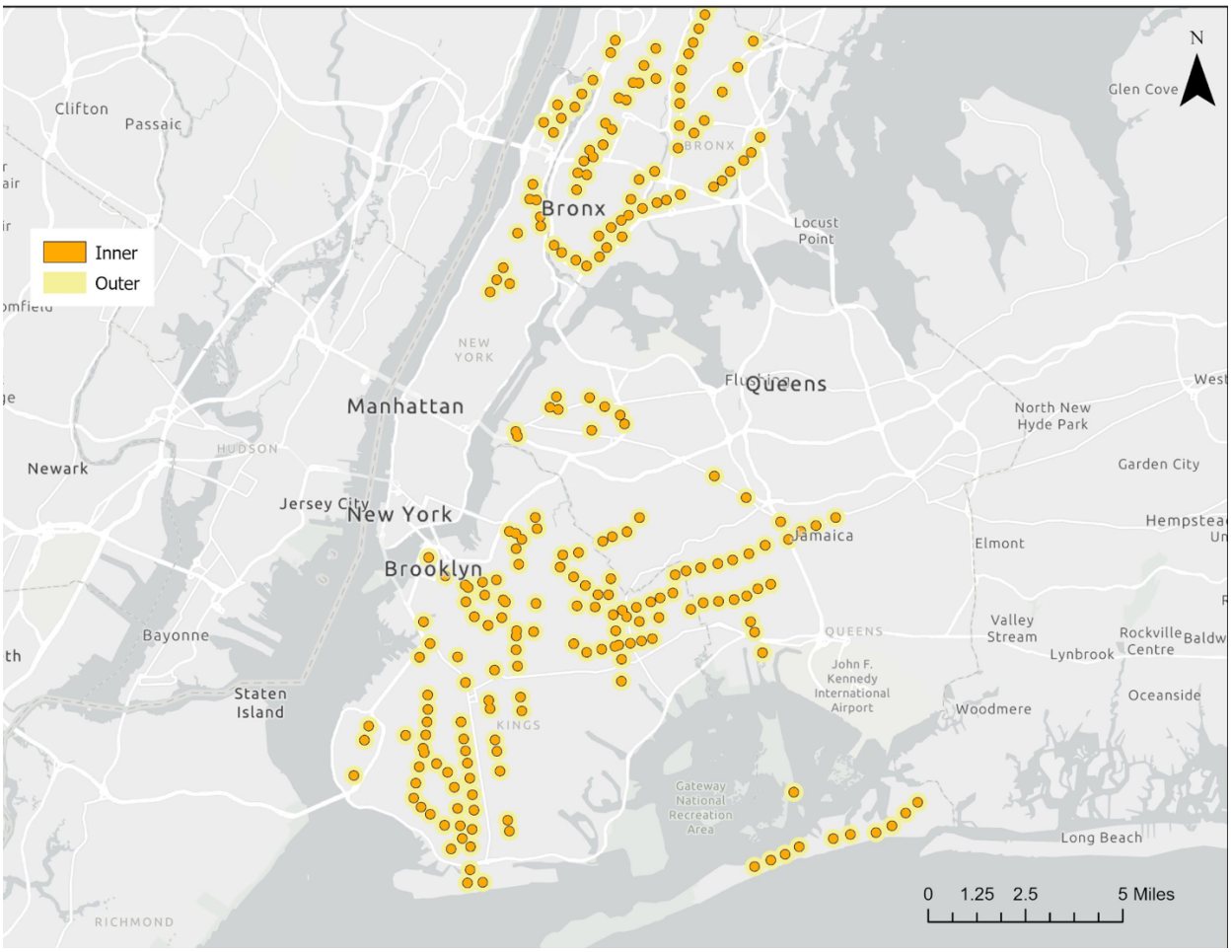


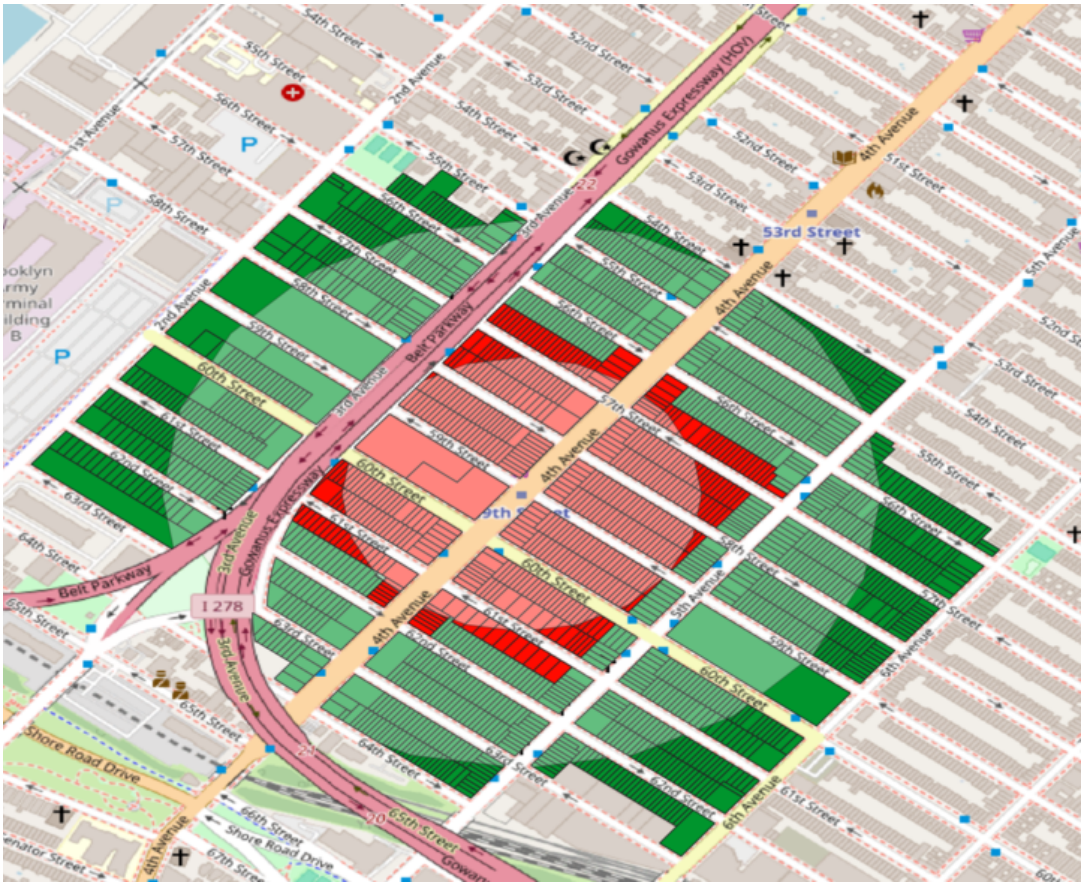
Figure 4. Inner and outer zones of TODX

Conclusion:

The addition of TODX zones to MTA station areas will encourage dense, sustainable and efficient development in parts of the city that have long been intended for intense use, despite current built environments. Although TODX is contextual to each area, if we assume that the new bonuses allow for R6-esque development, we can see that the TODX quarter-mile radii will increase R6 land by 22 sq. mi, or 41%. TODX will enable aggressive, yet contextual development to address our affordability crisis, and the vast scope spread across the entire city will mitigate the risk of intense gentrification in any neighborhood.

Appendix

1. Example Station Area: 59 St (N, R) in Brooklyn



In the above example, we can see the actual quarter mile zone (dark green), and the actual eighth mile zone (dark red) with the intersecting block treatments. Overlaid in light green and light red is what the zones would look like as drawn, if we didn't apply the block inclusion principle.

2. MIH Options

	Options	Affordable Floor Area Required +5% when off-site	Weighted Average of Income Bands	Limitations
Option 1 or Option 2 will always be made available	Option 1	25% of residential floor area	60% Area Median Income (AMI) or lower	No more than 3 income bands 10% floor area at 40% AMI No income band above 130% AMI
	Option 2	30% of residential floor area	80% AMI or lower	No more than 3 income bands No income band above 130% AMI
Options 3 and/or Option 4 may be made available	Option 3	20% of residential floor area	40% AMI or lower	No more than 3 income bands No income band above 130% AMI No public funding without HPD approval
	Option 4	30% of residential floor area	115% AMI or lower	No more than 4 income bands 5% floor area at 70% AMI, 5% floor area at 90% AMI No income band above 135% AMI No public funding Not available in Manhattan south of 96th Street

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